_eda_price

March 29, 2026

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import subprocess
import os
from scipy.stats import chi2_contingency
from datetime import timedelta

from etl.customer import aggregate_customer
from etl.subsidy import report_customer_conversion_by_subsidy_issue_status,␣
↪report_customer_metrics_by_subsidy_issue_status
from etl.subsidy import visualize_conversion_by_subsidy_issues,␣
↪report_customer_subsidy_issues_by_product
from etl.subsidy import visualize_customer_subsidy_issues_by_product,␣
↪visualize_policy_suspension_impact
from etl.price import visualize_conversion_by_price_sweet_spots,␣
↪visualize_conversion_by_price_sweet_spots_variant
from etl.price import report_product_conversion_by_price_quartile,␣
↪visualize_conversion_by_equipment_category_price
from etl.price import visualise_conversion_by_brand_price,␣
↪show_two_panel_conversion_by_brand_price
from etl.price import visualize_conversion_by_equipment_type_brand_price,␣
↪show_conversion_by_price_dashboard,␣
↪show_product_type_price_conversion_heatmap
from etl.season import visualize_conversion_by_season,␣
↪visualize_regional_seasonality
from etl.period import visualize_conversion_by_year,␣
↪report_best_and_worst_months, show_conversion_by_price_over_time,␣
↪overlay_all_years_on_one_plot
from etl.period import visualize_conversion_by_prices_sweet_spot_shift,␣
↪visualize_decision_times, show_decision_time_dashboard
from etl.product import visualise_heat_pump_performance,␣
↪show_product_type_by_brand_heatmap, show_product_type_by_agency_heatmap,␣
↪show_brand_by_product_type_heatmap
```

```python
import warnings
warnings.filterwarnings('ignore')

df_clean = pd.read_csv('cleaned_quote_data.csv')
df_clean['dt_creation_devis'] = pd.to_datetime(df_clean['dt_creation_devis'])
print(f"Loaded data: {len(df_clean):,} quotes from {df_clean['numero_compte'].
 ↪nunique():,} customers")
```

```
Loaded data: 37,008 quotes from 25,716 customers
```

```python
#customers = aggregate_customer(df_clean)
```

```python
customers = pd.read_csv('customer_master_data.csv')
```

```python
conv_no_issue, conv_with_issue =␣
 ↪report_customer_conversion_by_subsidy_issue_status(customers)
report_customer_metrics_by_subsidy_issue_status(customers)
visualize_conversion_by_subsidy_issues(customers, conv_no_issue,␣
 ↪conv_with_issue)
visualize_policy_suspension_impact(customers, price_var='max_out_of_pocket')

top_products, no_issue_pct, issue_pct =␣
 ↪report_customer_subsidy_issues_by_product(customers)
visualize_customer_subsidy_issues_by_product(customers, top_products,␣
 ↪no_issue_pct, issue_pct)

visualize_conversion_by_price_sweet_spots(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_conversion_by_price_sweet_spots_variant(customers,␣
 ↪price_var='max_out_of_pocket')
report_product_conversion_by_price_quartile(customers)
visualize_conversion_by_equipment_category_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualise_conversion_by_brand_price(customers, price_var='max_out_of_pocket')
show_two_panel_conversion_by_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_conversion_by_equipment_type_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')

visualize_conversion_by_season(customers, price_var='max_out_of_pocket')
visualize_regional_seasonality(customers, price_var='max_out_of_pocket')

visualize_conversion_by_year(customers, price_var='max_out_of_pocket')
report_best_and_worst_months(customers, price_var='max_out_of_pocket')
show_conversion_by_price_over_time(customers, price_var='max_out_of_pocket')
overlay_all_years_on_one_plot(customers, price_var='max_out_of_pocket')
```

```
visualize_conversion_by_prices_sweet_spot_shift(customers,␣
↪price_var='max_out_of_pocket')
visualize_decision_times(customers, price_var='max_out_of_pocket')
show_product_type_price_conversion_heatmap(customers,␣
↪price_var='max_out_of_pocket')
show_decision_time_dashboard(customers, price_var='max_out_of_pocket')

visualise_heat_pump_performance(customers, price_var='max_out_of_pocket')

show_conversion_by_price_dashboard(customers, price_var='max_out_of_pocket')

show_product_type_by_brand_heatmap(customers)
show_product_type_by_agency_heatmap(customers)
show_brand_by_product_type_heatmap(customers)
```

```
Customer conversion by subsidy issue status:

                   conversion_rate  customer_count  converted_count
had_subsidy_issue
False                     0.362042           17335           6276.0
True                      0.485579            6553           3182.0

Chi-square p-value: 7.601147e-68
Statistically significant: YES

Conversion without issues: 36.20%
Conversion with issues: 48.56%
Difference: 12.4 percentage points
Customer-level finding: +12.4 points

================================================================================
Customer characteristics by subsidy issue status
================================================================================

Metric comparison:

total_quotes:
  No issues: 1.38
  With issues: 1.53
  Ratio: 1.11x

avg_quote_amount:
  No issues: 6164.43
  With issues: 9935.00
  Ratio: 1.61x
```

```
max_quote_amount:
  No issues: 6571.38
  With issues: 10832.75
  Ratio: 1.65x

customer_duration_days:
  No issues: 17.87
  With issues: 24.31
  Ratio: 1.36x

price_range:
  No issues: 868.91
  With issues: 1702.03
  Ratio: 1.96x

multiple_quotes:
  No issues: 0.28
  With issues: 0.36
  Ratio: 1.28x

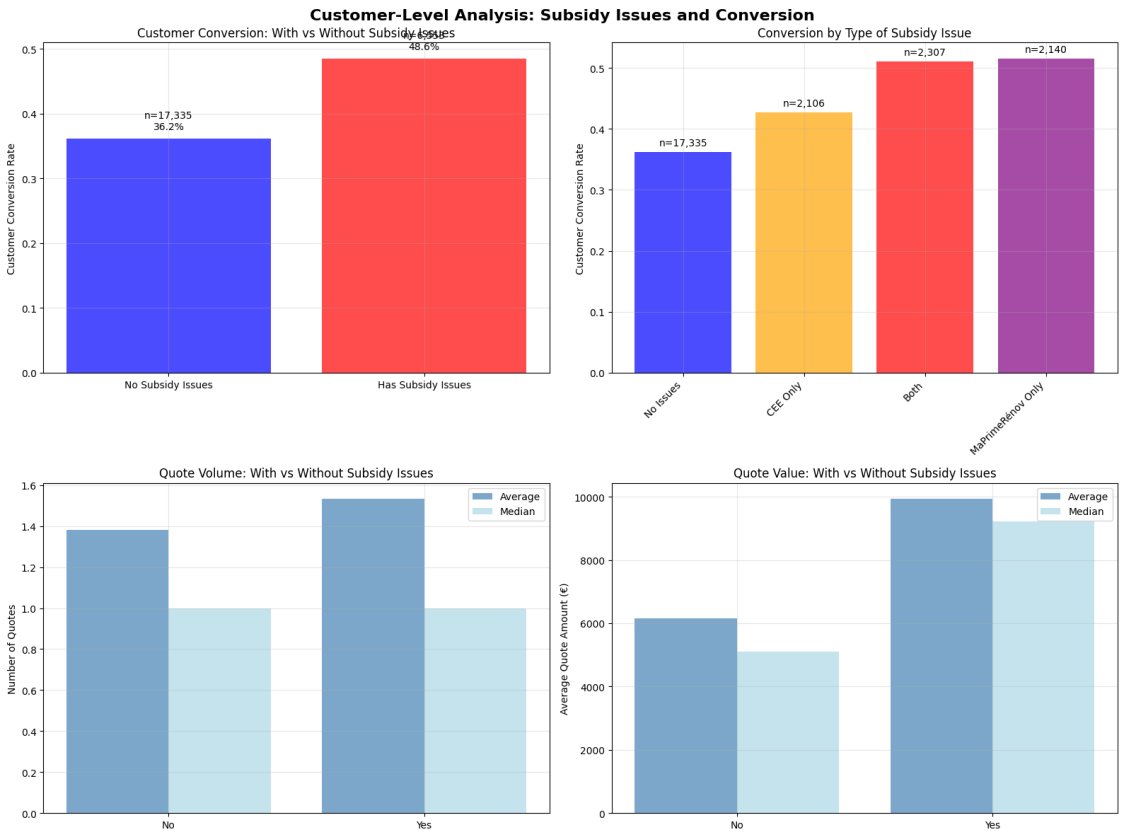
Quote pattern distribution by subsidy issue:
had_subsidy_issue  False  True
quote_pattern
Single quote       0.722  0.644
Three+ quotes      0.066  0.111
Two quotes         0.212  0.245

Decision speed by subsidy issue:
had_subsidy_issue   False  True
decision_speed
Long (>30 days)     0.070  0.101
Medium (8-30 days)  0.031  0.047
Quick (2-7 days)    0.040  0.063
Same day decision   0.858  0.788

Product family by subsidy issue status:
had_subsidy_issue                False  True
main_product_family
Appareil hybride                 0.002  0.017
Autres                           0.013  0.007
Chaudière                        0.398  0.082
Climatisation                    0.242  0.058
ECS : Chauffe-eau ou adoucisseur 0.054  0.020
Emetteur de chauffage  ou chappe 0.004  0.003
Fumisterie                       0.001  0.000
Photovoltaïque                   0.027  0.013
Plomberie Sanitaire              0.006  0.000
Pompe à chaleur                  0.076  0.423
```

```
Poêle                              0.172  0.374
Produit VMC                        0.005  0.000
Unknown                            0.001  0.002

Conversion by subsidy issue type:
                        mean  count
subsidy_issue_type
MaPrimeRénov Only   0.515888   2140
Both                0.510620   2307
CEE Only            0.427350   2106
No Issues           0.362042  17335
```

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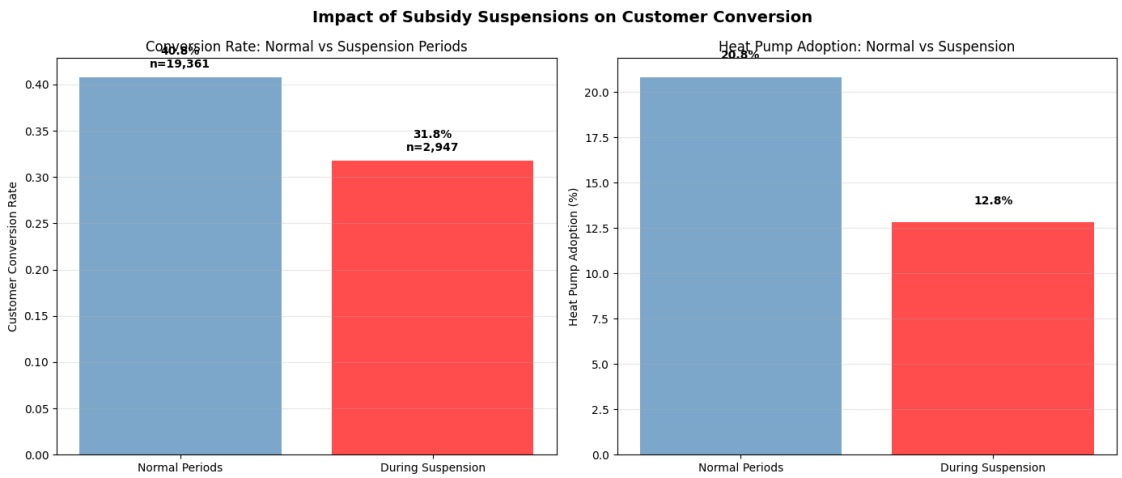
================================================================================
Policy Suspension Impact
================================================================================

Analyzing 22,308 customers
Price range: €1,339 to €28,212
Median price: €6,321
Unique values in during_suspension: [False  True]
```

```
Conversion during subsidy suspensions:
                      mean  count
Normal Periods     0.40814  19361
During Suspension  0.31795   2947

Suspension impact p-value: 0.0000
Statistically significant: YES

Product mix during suspensions:
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove  Unknown
during_suspension
False                    17.3    31.7       18.8    8.1   23.8      0.2
True                     38.0    28.1       10.6    6.2   17.1      0.0
```

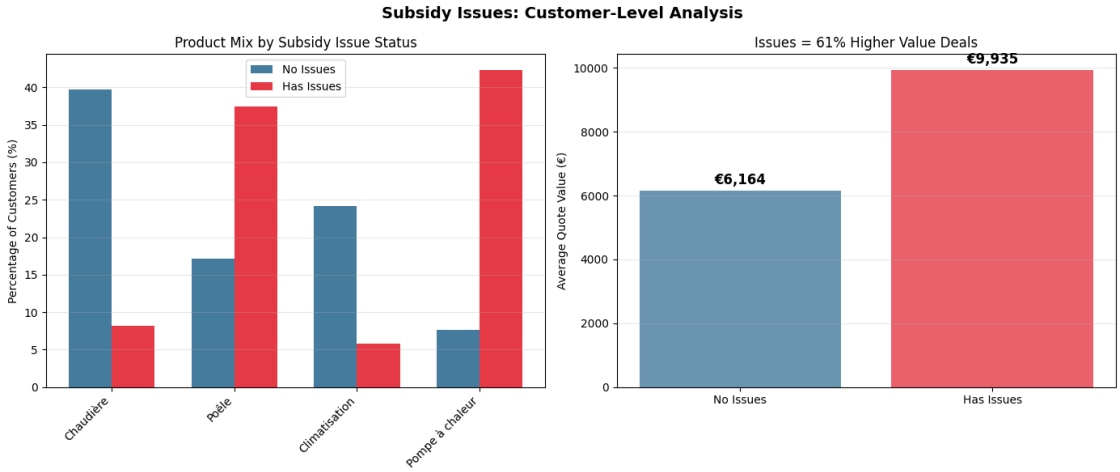



```
Unique values in main_product_family:
main_product_family
Chaudière                          7430
Poêle                              5431
Climatisation                      4573
Pompe à chaleur                    4090
ECS : Chauffe-eau ou adoucisseur   1060
Photovoltaïque                      549
Autres                              270
Appareil hybride                    151
Plomberie Sanitaire                 103
Emetteur de chauffage  ou chappe     93
Produit VMC                          89
Unknown                              36
Fumisterie                           13
Name: count, dtype: int64
```

```
Top products:
main_product_family
Chaudière                          7430
Poêle                              5431
Climatisation                      4573
Pompe à chaleur                    4090
ECS : Chauffe-eau ou adoucisseur   1060
Photovoltaïque                      549
Autres                              270
Appareil hybride                    151
Plomberie Sanitaire                 103
Emetteur de chauffage  ou chappe     93
Name: count, dtype: int64

Using these products: ['Chaudière', 'Poêle', 'Climatisation', 'Pompe à chaleur']

No Issue %: [np.float64(39.76348428035766), np.float64(17.184886068647245),
np.float64(24.176521488318432), np.float64(7.597346408999135)]
Has Issue %: [np.float64(8.194719975583702), np.float64(37.41797649931329),
np.float64(5.829391118571647), np.float64(42.316496261254386)]
```

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================================================================================
Threshold effect
================================================================================

Analyzing 22,308 customers
Price range: €1,339 to €28,212
Median price: €6,321
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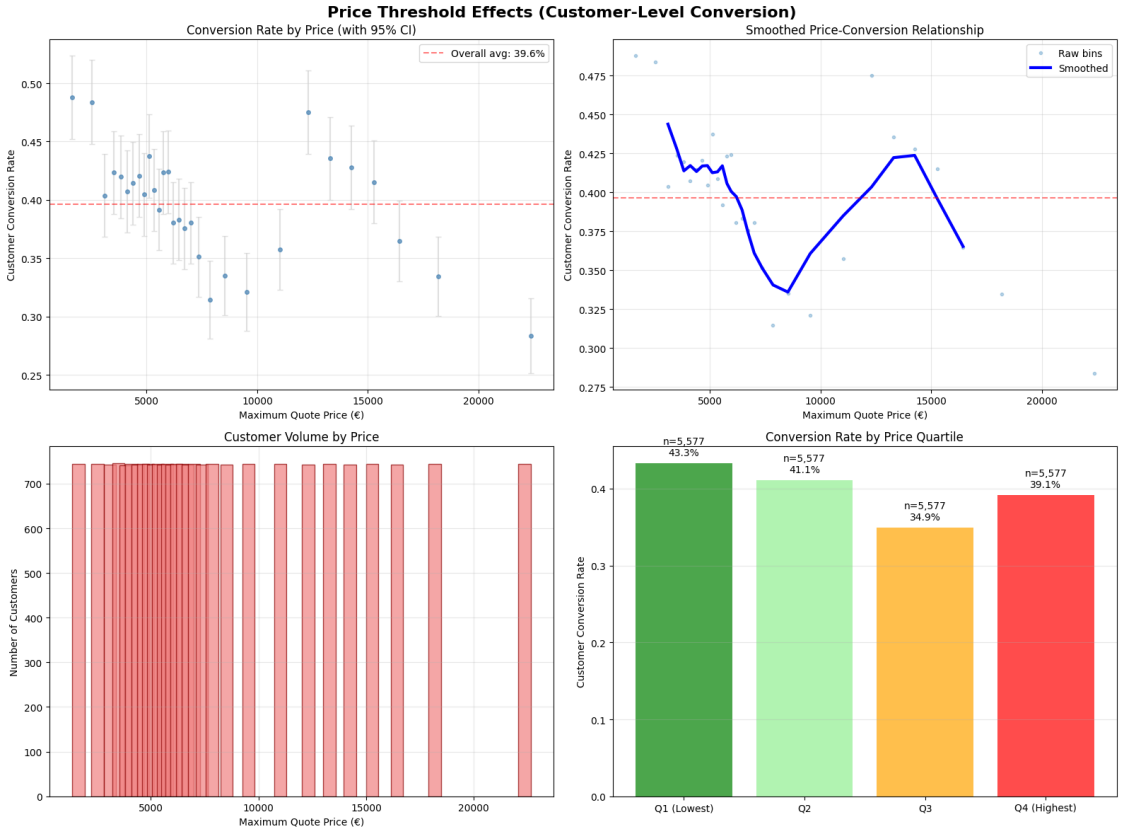


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================================================================================
THRESHOLD EFFECT: KEY FINDINGS
================================================================================

Conversion by Price Quartile:
  Q1 (Lowest): 43.3% (n=5,577)
  Q2: 41.1% (n=5,577)
  Q3: 34.9% (n=5,577)
  Q4 (Highest): 39.1% (n=5,577)

Difference between lowest and highest quartile: 4.2 percentage points

Sweet spots (above-average conversion):
  €1,643: 48.8% (n=744)
  €2,523: 48.4% (n=744)
  €3,102: 40.4% (n=743)
  €3,506: 42.4% (n=746)
  €3,822: 42.0% (n=741)
  €4,110: 40.7% (n=744)
  €4,393: 41.4% (n=743)
```

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  €4,654: 42.1% (n=744)
  €4,884: 40.5% (n=744)
  €5,107: 43.7% (n=743)
  €5,341: 40.9% (n=744)
  €5,769: 42.3% (n=744)
  €5,963: 42.4% (n=743)
  €12,305: 47.5% (n=743)
  €13,300: 43.5% (n=744)
  €14,250: 42.8% (n=743)
  €15,283: 41.5% (n=744)

Analyzing 22,308 customers
Price range: €1,339 to €28,212
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================================================================================
Product mix by price quartile
================================================================================

Product distribution by price quartile (%):
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove  Unknown
price_quartile
Q1_Low                   24.6    44.2        1.7   18.7   10.5      0.1
Q2_MidLow                14.8    36.2        1.1    2.4   45.4      0.1
Q3_MidHigh               25.3    31.4        5.3    5.8   32.2      0.1
Q4_High                  14.9    12.2       63.7    7.1    1.9      0.2

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Conversion rates by product and price quartile
================================================================================

Heat Pump (n=4,385):
  Q1_Low: 49.0%
  Q2_MidLow: 40.0%
  Q3_MidHigh: 53.5%
  Q4_High: 42.0%

Boiler (n=7,369):
  Q1_Low: 46.4%
  Q2_MidLow: 40.2%
  Q3_MidHigh: 38.7%
  Q4_High: 36.3%

AC (n=4,657):
  Q1_Low: 36.9%
  Q2_MidLow: 30.2%
  Q3_MidHigh: 28.0%
  Q4_High: 31.8%
```

```
Stove (n=5,427):
  Q1_Low: 48.7%
  Q2_MidLow: 45.6%
  Q3_MidHigh: 34.8%
  Q4_High: 34.9%

================================================================================
HEAT PUMP CUSTOMERS (n=4,875)
================================================================================
Average price: €15,646
Conversion rate: 42.6%
Subsidy issue rate: 68.1%

Heat pump adoption by quartile:
  Q1_Low: 1.8%
  Q2_MidLow: 1.2%
  Q3_MidHigh: 6.0%
  Q4_High: 71.2%

================================================================================
Equipment Category Segment Effect
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Analyzing 22,308 customers
Price range: €1,339 to €28,212
Median price: €6,321
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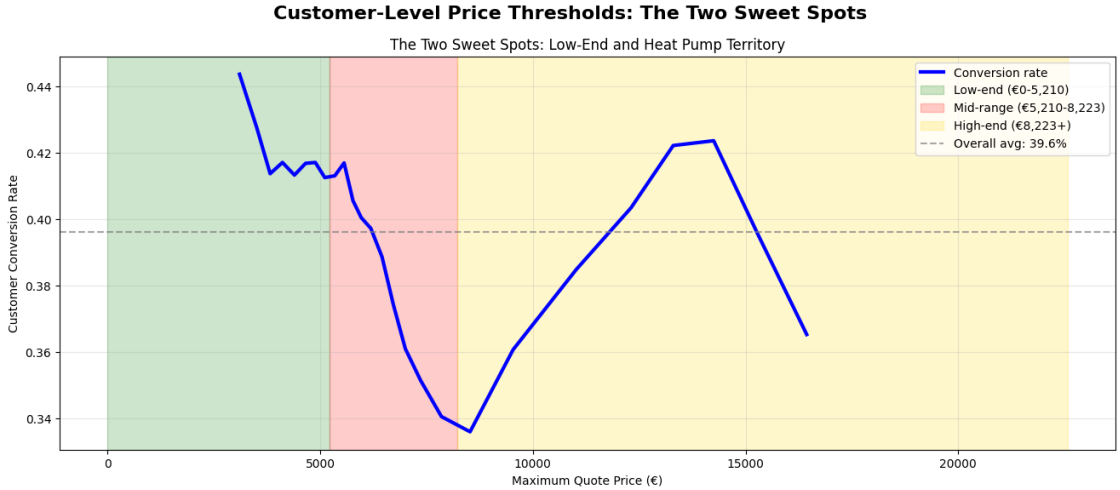

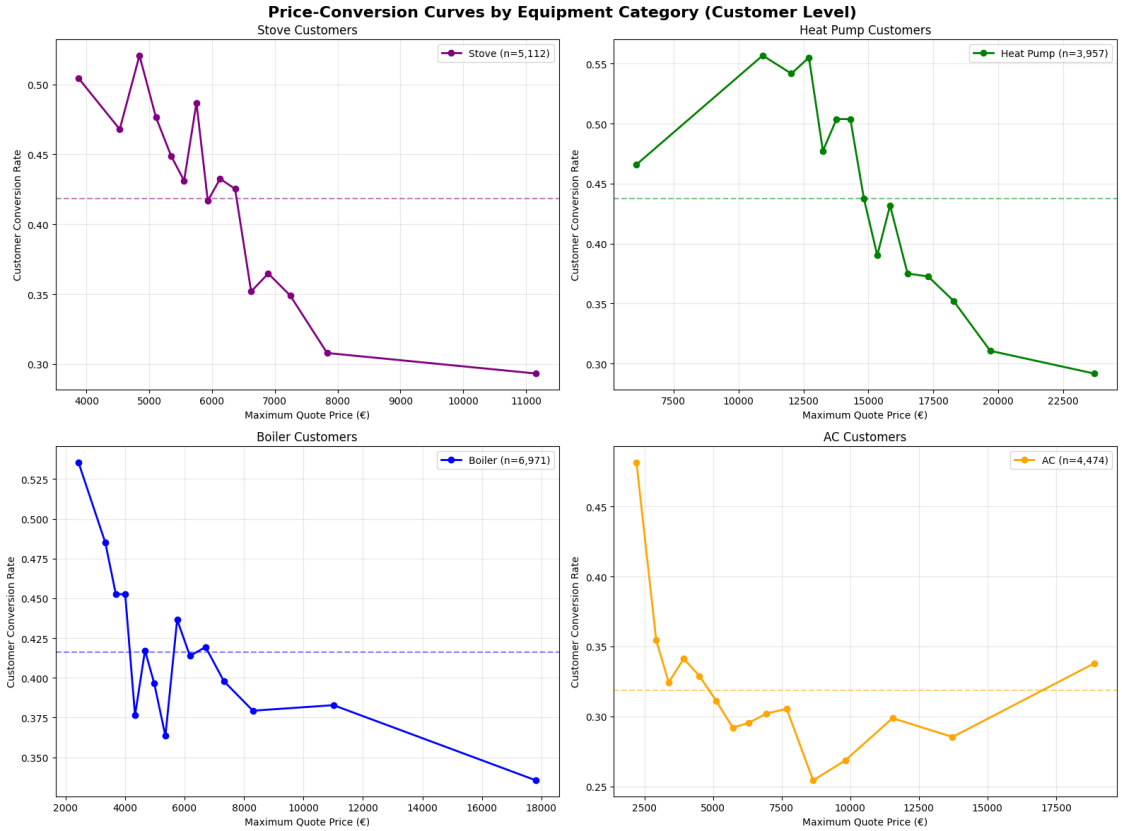


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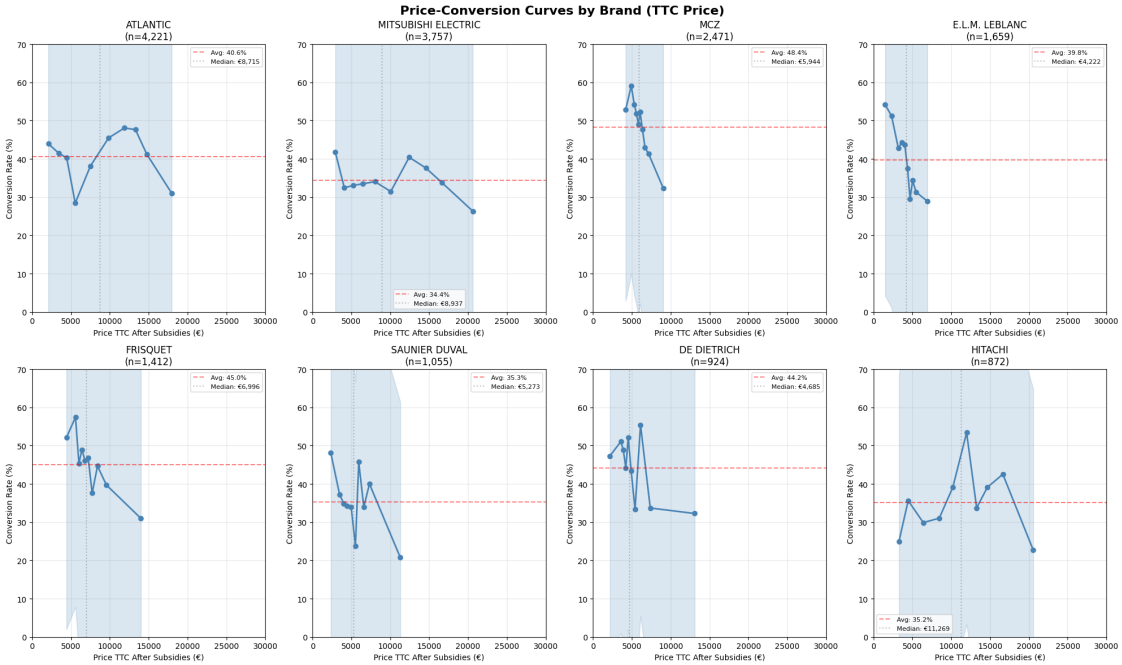
================================================================================
Conversion by Brand Price
================================================================================

Analyzing 22,308 customers
Price range: €1,339 to €28,212
Median price: €6,321
Top 8 brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC', 'MCZ', 'E.L.M. LEBLANC',
'FRISQUET', 'SAUNIER DUVAL', 'DE DIETRICH', 'HITACHI']
```

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Analyzing 22,308 customers
Price range: €1,339 to €28,212
Median price: €6,321
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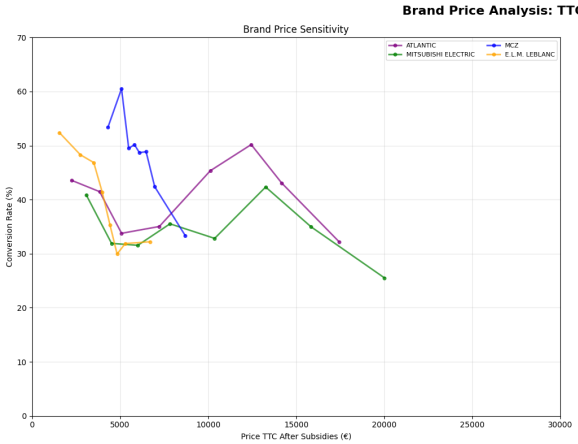

Brand Performance Summary

Brand	Customers	Avg Conv	Avg Price	Best Price Range	Best Conv
ATLANTIC	4,221	40.6%	€9,072	€10,953-13,967	47.9%
MITSUBISHI ELECTRIC	3,757	34.4%	€10,115	€11,289-15,591	39.0%
MCZ	2,471	48.4%	€6,115	€1,800-5,144	56.0%
E.L.M. LEBLANC	1,659	39.8%	€4,127	€1,328-2,845	52.7%
FRISQUET	1,412	45.0%	€7,613	€2,391-5,812	54.8%
SAUNIER DUVAL	1,055	35.3%	€5,593	€1,343-3,774	42.7%
DE DIETRICH	924	44.2%	€5,533	€1,341-3,750	49.2%
HITACHI	872	35.2%	€10,998	€9,190-12,676	46.3%

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================================================================================
TTC Price-conversion curves: 4 equipment types, multiple brands per graph
================================================================================

Analyzing 22,308 customers
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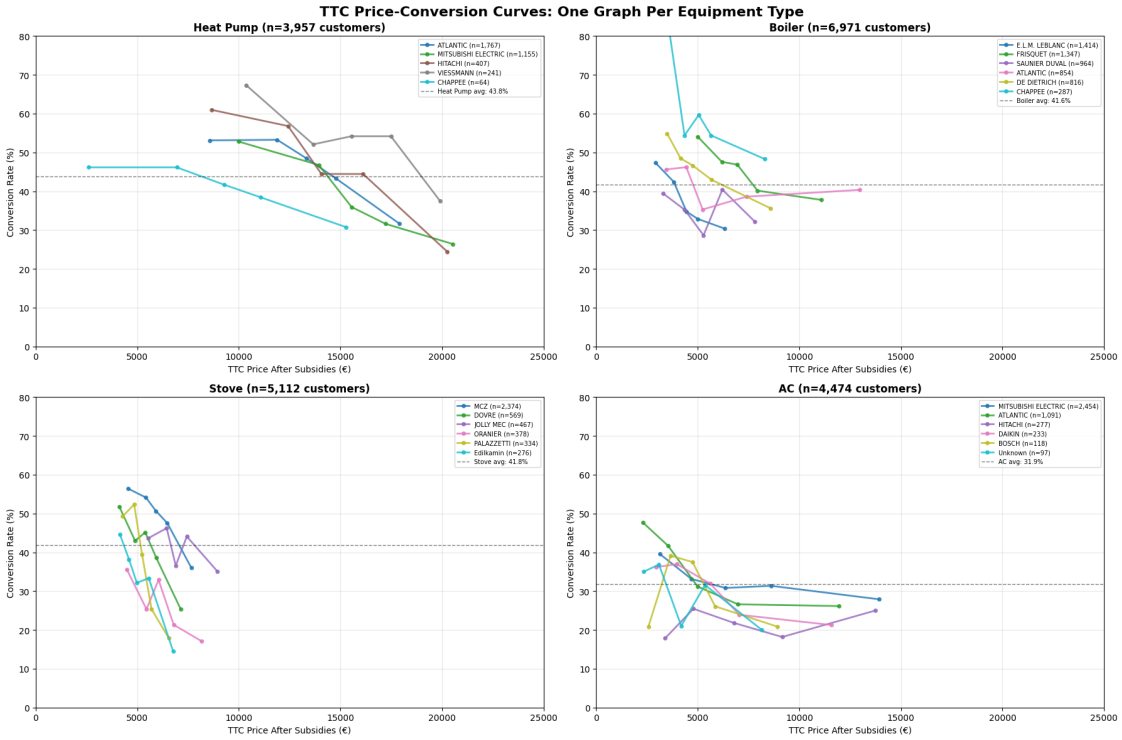
```
Price range: €1,339 to €28,212
Median price: €6,321

Heat Pump - Top brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC', 'HITACHI',
'VIESSMANN', 'CHAPPEE']

Boiler - Top brands: ['E.L.M. LEBLANC', 'FRISQUET', 'SAUNIER DUVAL', 'ATLANTIC',
'DE DIETRICH', 'CHAPPEE']

Stove - Top brands: ['MCZ', 'DOVRE', 'JOLLY MEC', 'ORANIER', 'PALAZZETTI',
'Edilkamin']

AC - Top brands: ['MITSUBISHI ELECTRIC', 'ATLANTIC', 'HITACHI', 'DAIKIN',
'BOSCH', 'Unknown']
```

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================================================================================
Seasonal Effects on Customer Conversion
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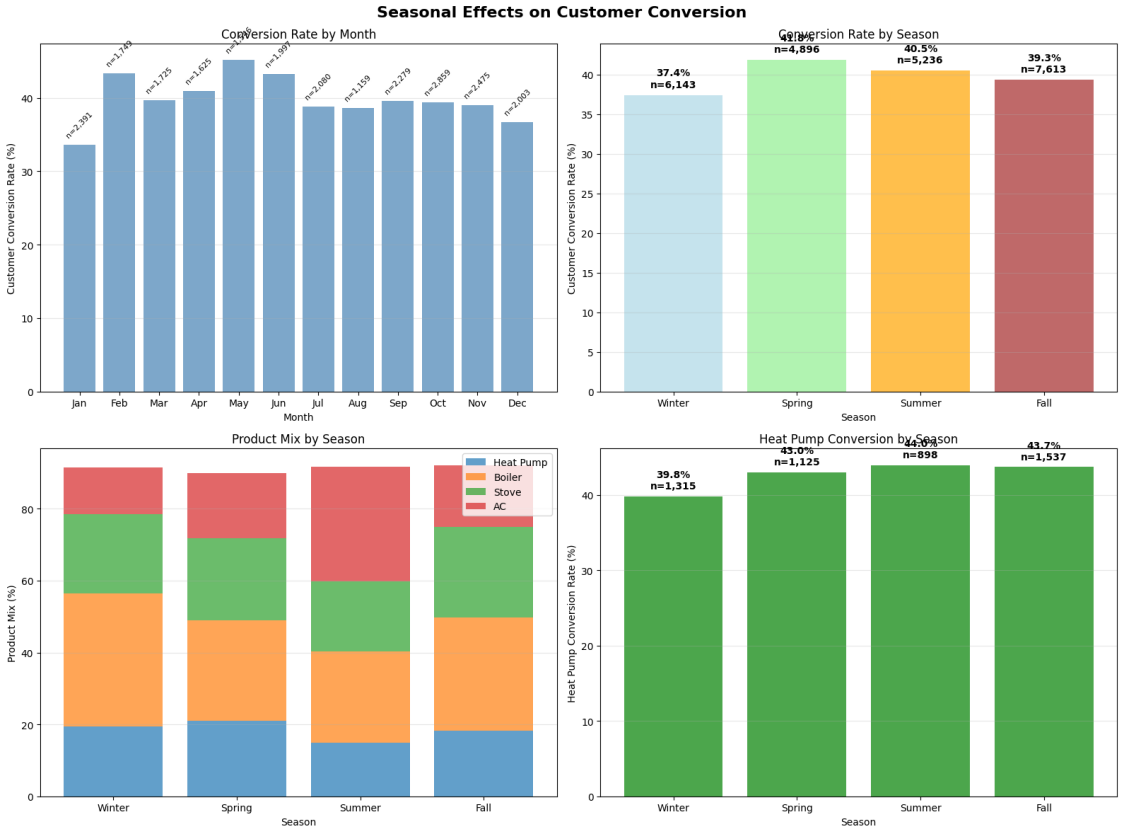


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================================================================================
SEASONAL STATISTICS
================================================================================

Conversion by season:
  Winter: 37.4% (n=6,143)
  Spring: 41.8% (n=4,896)
  Summer: 40.5% (n=5,236)
  Fall: 39.3% (n=7,613)

Seasonality chi-square p-value: 0.0000
Statistically significant: YES

================================================================================
Regional Seasonal Pattern
================================================================================

Analyzing 22,308 customers
Price range: €1,339 to €28,212
Median price: €6,321
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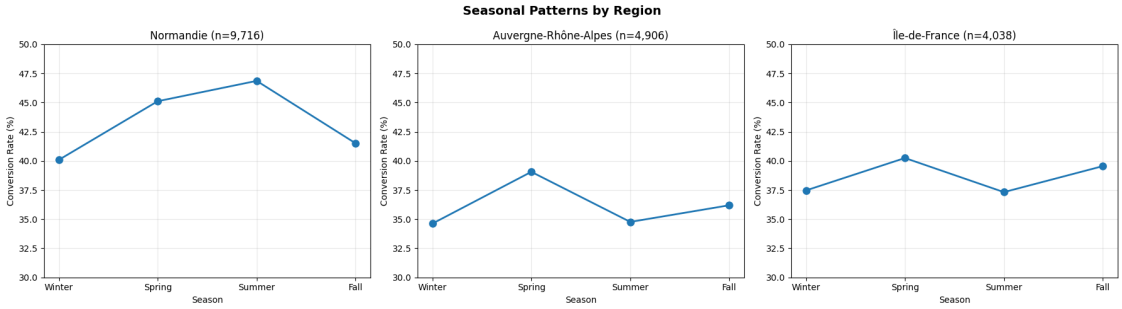
Top regions: ['Normandie', 'Auvergne-Rhône-Alpes', 'Île-de-France']
```

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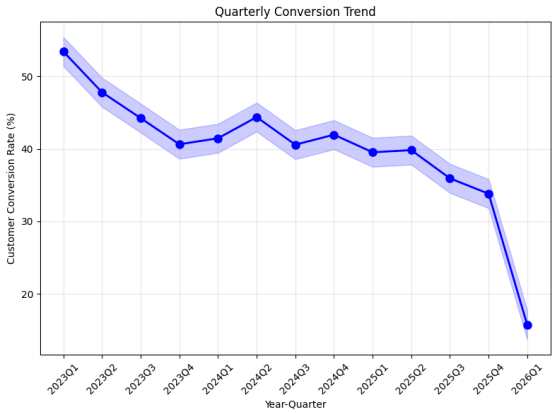
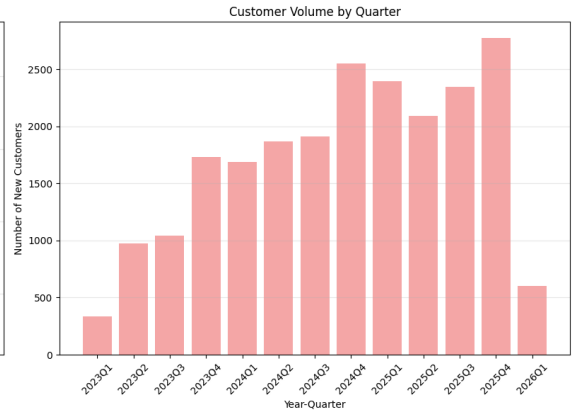
================================================================================
Year-over-year trends
================================================================================

Analyzing 22,308 customers
Price range: €1,339 to €28,212
Median price: €6,321

Quarterly conversion rates:
                  mean  count
year_quarter
2023Q1        0.534125    337
2023Q2        0.477994    977
2023Q3        0.442418   1042
2023Q4        0.406358   1730
2024Q1        0.414591   1686
2024Q2        0.443790   1868
2024Q3        0.405759   1910
2024Q4        0.419545   2548
2025Q1        0.395329   2398
2025Q2        0.398277   2089
2025Q3        0.359368   2343
2025Q4        0.338256   2776
2026Q1        0.157285    604
```



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================================================================================
Best and Worst Months Report
================================================================================

Analyzing 22,308 customers
Price range: €1,339 to €28,212
Median price: €6,321

Conversion during subsidy suspensions:
                      mean  count
Normal Periods     0.40814  19361
During Suspension  0.31795   2947

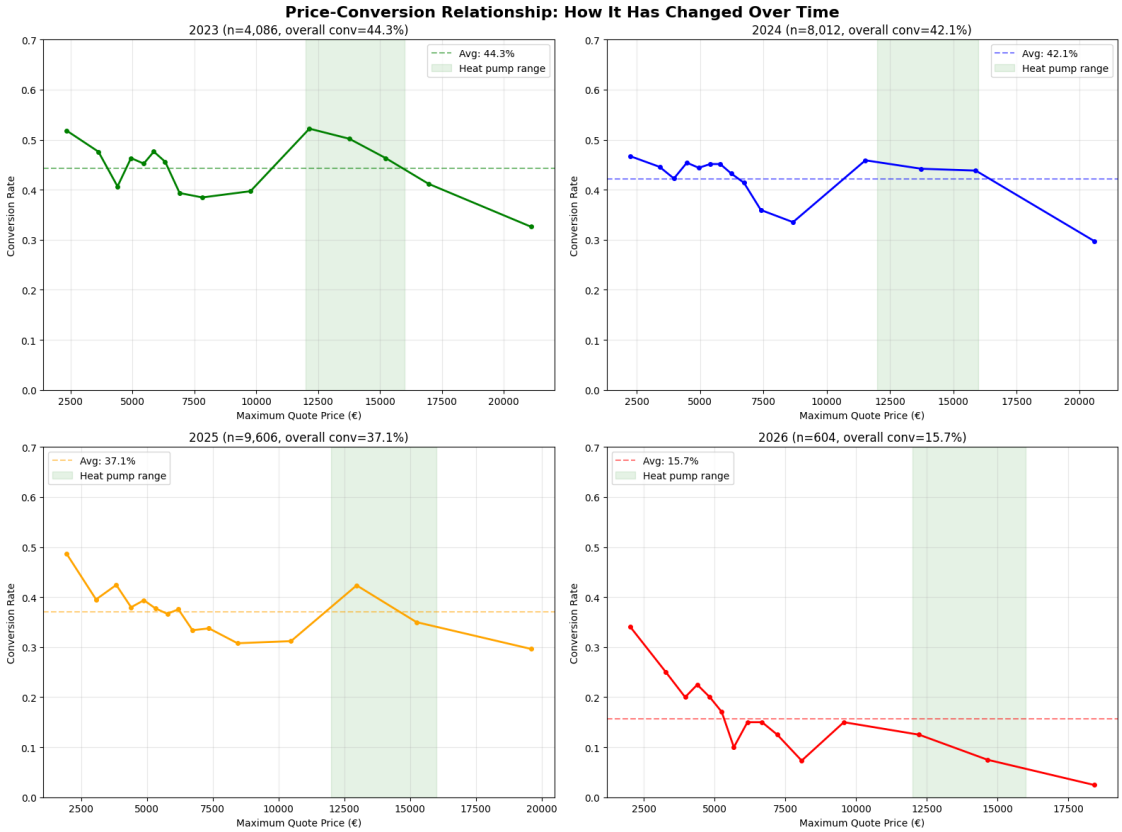
Suspension impact p-value: 0.0000
Statistically significant: YES

Best month: 5 (44.9%, n=1478)
Worst month: 1 (32.6%, n=2055)

Best season: Spring (42.0%)
Worst season: Winter (37.3%)

Heat pump conversion by season:
  Winter: 41.1%
  Spring: 44.0%
  Summer: 45.5%
  Fall: 44.4%

Analyzing 22,308 customers
Price range: €1,339 to €28,212
Median price: €6,321
```

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Analyzing 22,308 customers
Price range: €1,339 to €28,212
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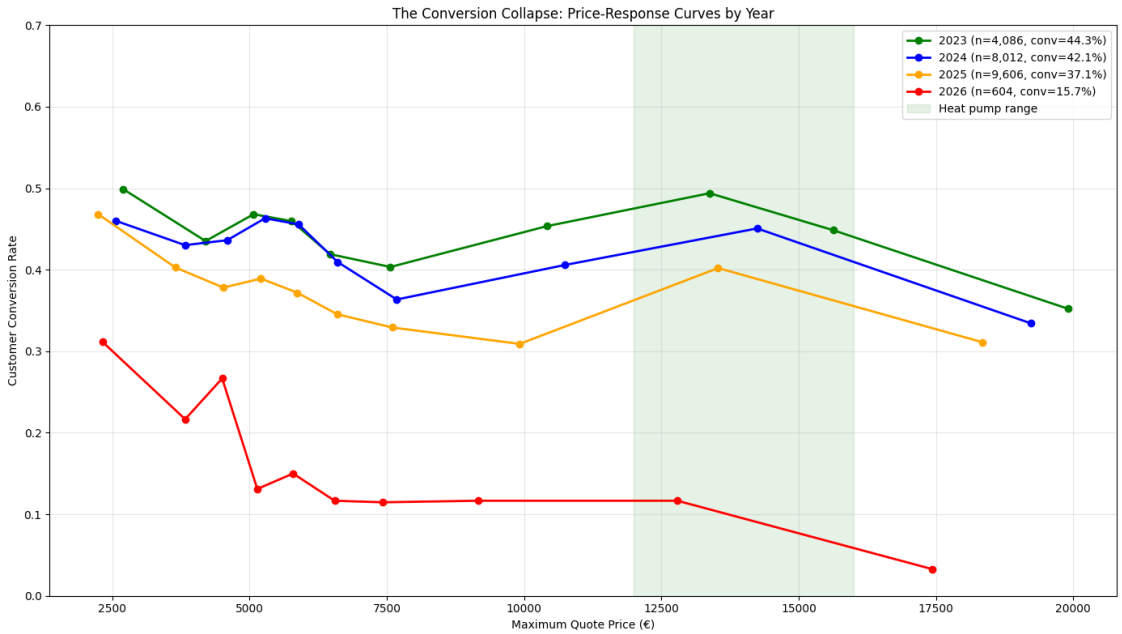


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================================================================================
Sweet Spot Analysis Over Time
================================================================================

Analyzing 22,308 customers
Price range: €1,339 to €28,212
Median price: €6,321

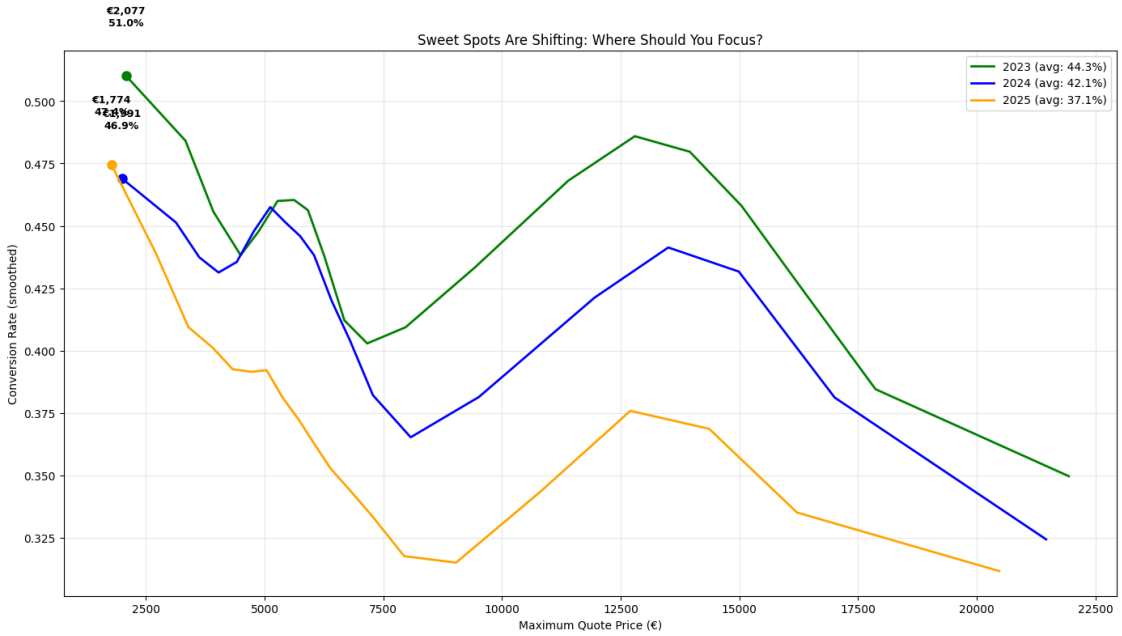
Sweet spots by year (price points with conversion > avg + 3%):

2023 (overall avg: 44.3%):
  €2,077: 53.7% (n=205)
  €3,920: 48.1% (n=210)
  €11,395: 49.5% (n=204)
  €12,798: 49.0% (n=204)
  €13,954: 49.8% (n=205)

2024 (overall avg: 42.1%):
  €1,991: 48.9% (n=401)
  €5,118: 49.5% (n=400)
  €6,041: 46.9% (n=401)
  €11,950: 47.8% (n=400)
  €14,988: 48.2% (n=400)

2025 (overall avg: 37.1%):
  €1,774: 50.3% (n=481)
```

```
€2,712: 43.3% (n=480)
€3,910: 43.3% (n=480)
€5,041: 42.1% (n=480)
€12,704: 42.5% (n=480)
```

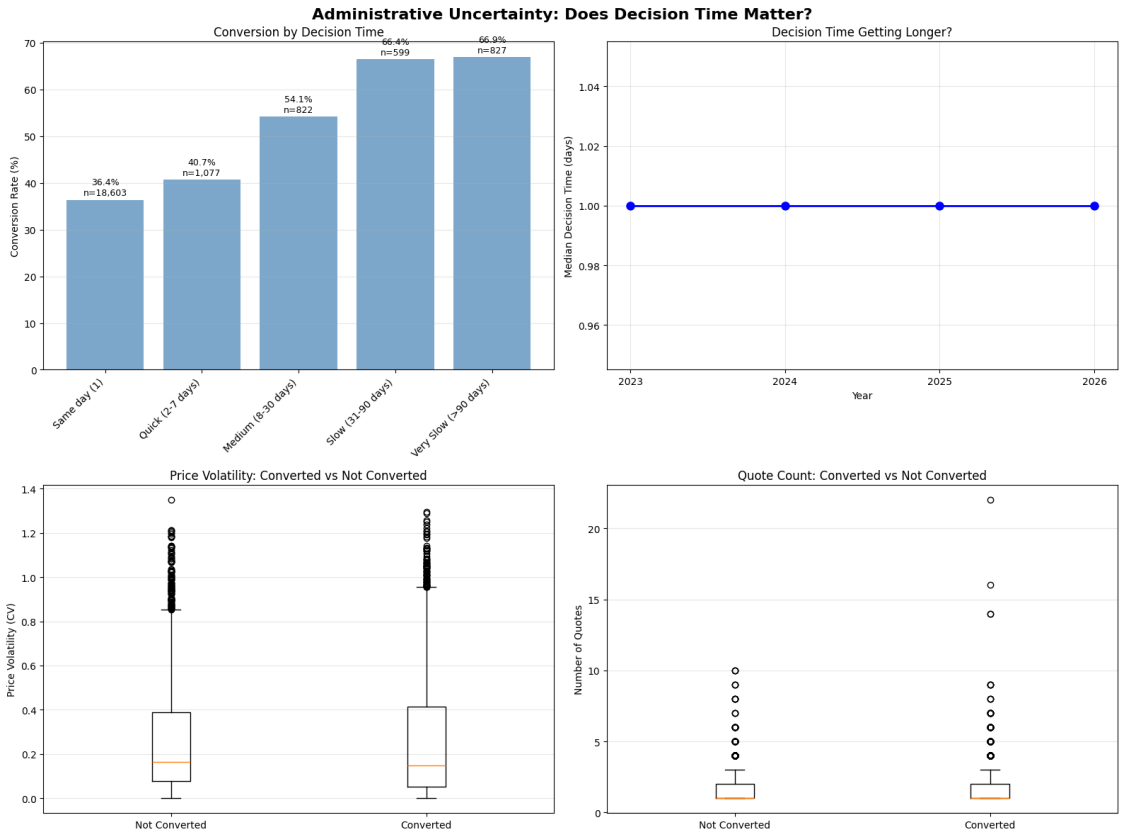



```
================================================================================
Administrative Uncertainty Effect
================================================================================

Analyzing 22,308 customers
Price range: €1,339 to €28,212
Median price: €6,321

Decision time distribution:
count    22308.000000
mean        20.289851
std         86.447456
min          1.000000
25%          1.000000
50%          1.000000
75%          1.000000
max       1082.000000
Name: decision_days, dtype: float64

Customers with 1-day decision: 18,603 (83.4%)
```

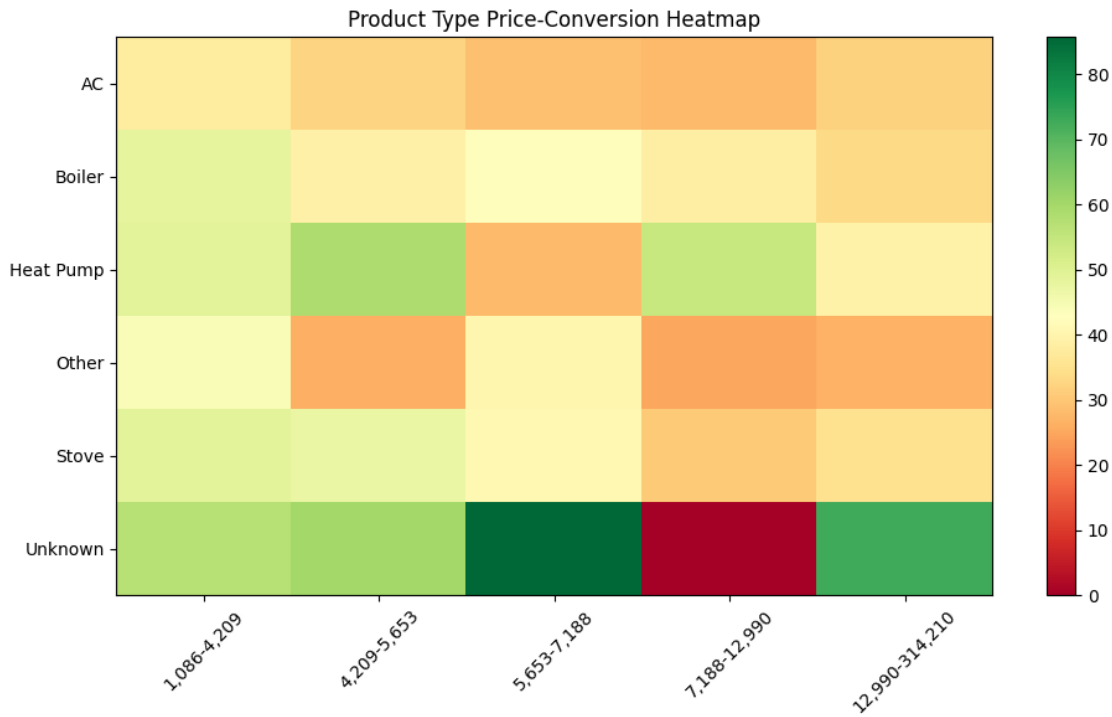



```
================================================================================
Product Type - Price Conversion Heatmap
================================================================================
Simple pivot table:
price_bin_simple         (1086.579, 4209.46]  (4209.46, 5653.482]  \
main_equipment_category
AC                                37.541806            32.258065
Boiler                            48.332478            38.848548
Heat Pump                         48.888889            58.536585
Other                             43.972999            26.262626
Stove                             49.090909            47.281993
Unknown                           57.142857            60.000000

price_bin_simple         (5653.482, 7188.172]  (7188.172, 12990.276]  \
main_equipment_category
AC                                29.127517             28.012279
Boiler                            42.458101             38.171141
Heat Pump                         28.000000             54.565953
Other                             40.277778             24.926686
Stove                             41.168353             30.582524
```

```
Unknown                                85.714286                0.000000

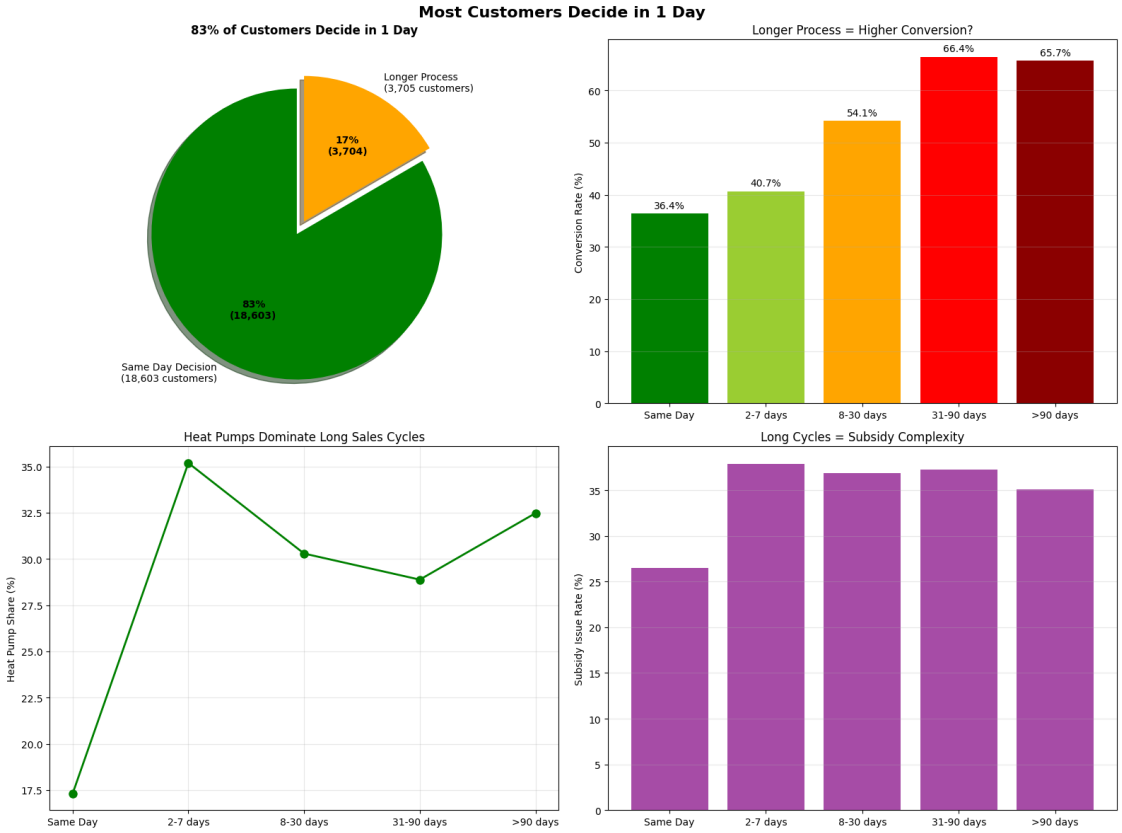
price_bin_simple          (12990.276, 314210.61]
main_equipment_category
AC                                      31.889081
Boiler                                  33.393178
Heat Pump                               39.489390
Other                                   26.602564
Stove                                   35.000000
Unknown                                 72.727273
```




```
================================================================================
CHECKING MAIN PRODUCT FAMILY
================================================================================
main_product_family
Chaudière                           7430
Poêle                               5431
Climatisation                       4573
Pompe à chaleur                     4090
ECS : Chauffe-eau ou adoucisseur    1060
Photovoltaïque                       549
Autres                               270
Appareil hybride                     151
```

```
Plomberie Sanitaire                    103
Emetteur de chauffage  ou chappe        93
Produit VMC                             89
Unknown                                 36
Fumisterie                              13
Name: count, dtype: int64

Analyzing 22,308 customers
Price range: €1,339 to €28,212
Median price: €6,321
```

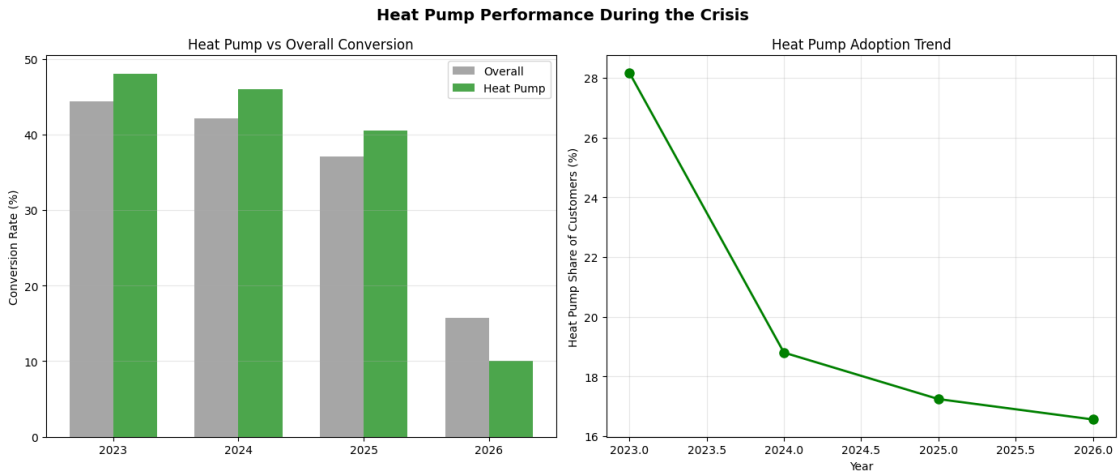



```
================================================================================
Heat Pump Performance Over Time
================================================================================

Analyzing 22,308 customers
Price range: €1,339 to €28,212
Median price: €6,321

Heat pump customers by year:
        conversion_rate  customer_count  avg_price  subsidy_issue_rate
```

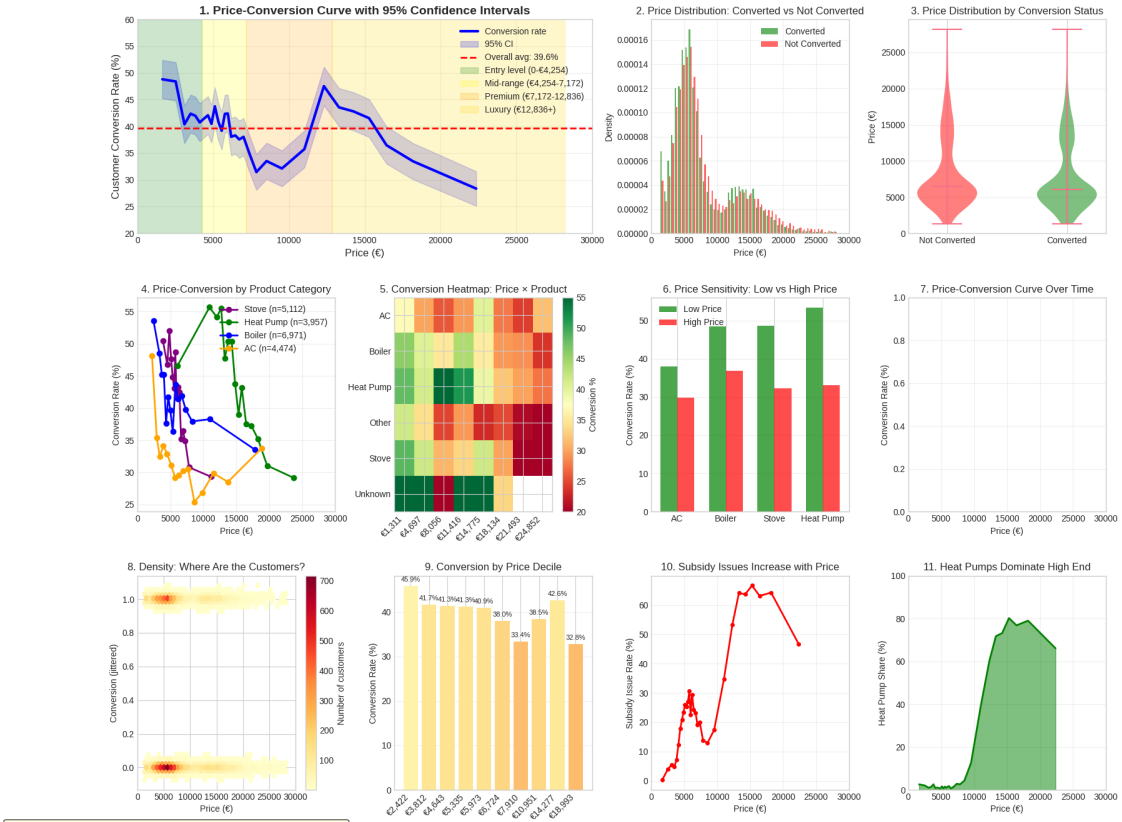
```
period
2023          0.480          1151  14760.547          0.700
2024          0.459          1506  15021.693          0.663
2025          0.405          1656  15145.143          0.754
2026          0.100           100  14783.915          0.750
```




```
Analyzing 22,308 customers
Price range: €1,339 to €28,212
Median price: €6,321

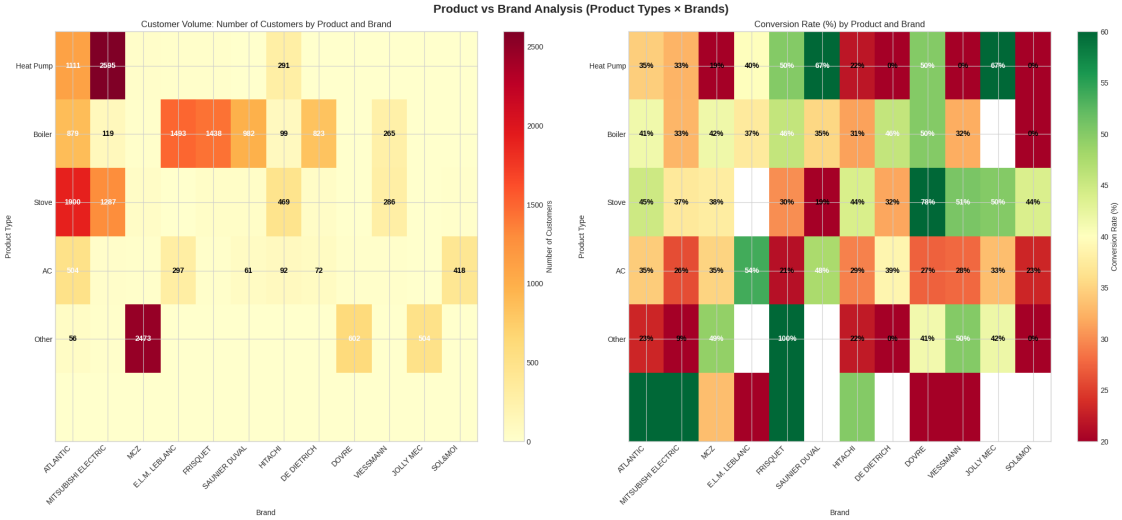
================================================================================
DYNAMIC INSIGHTS CALCULATED:
================================================================================
Sweet spots: €1.6k (48.8%) and €2.5k (48.4%)
Dead zone: €19.4k-28.2k (28-37%)
Heat pumps: 71% subsidy issues, 43.6% conversion
Price sensitivity: Heat Pump (20.3pts) > Stove (16.4pts) > Boiler (11.6pts) > AC
(8.2pts)
Same day decisions: 83%
```

Price-Conversion Dependency Dashboard



Key Insights:
- Two sweet spots: €1.6k (48.8%) and €2.5k (48.4%)
- Dead zone: €19.4k-28.2k (28-37%)
- Heat pumps: 71% subsidy issues, 43.6% conversion
- Price sensitivity: Heat Pump (20.3pts) > Stove (16.4pts) > Boiler (11.6pts) > AC (8.2pts)
- 83% decide in 1 day - price is not the only factor

Product vs Brand Analysis (Product Types × Brands)



```
================================================================================
PRODUCT × BRAND ANALYSIS
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC   MCZ  E.L.M. LEBLANC
FRISQUET  SAUNIER DUVAL  HITACHI  DE DIETRICH  DOVRE  VIESSMANN  JOLLY MEC
SOL&MOI
main_equipment_category
AC                            1111                 2595    21              15
16              9      291            5      2          3          3        1
Boiler                         879                  119    12            1493
1438            982       99          823      6        265          0        2
Heat Pump                     1900                 1287    45               0
40             32      469           41      9        286          2       16
Other                          504                   23    20             297
14             61       92           72     11         18          9      418
Stove                           56                   11  2473               0
1              0        9            3    602          2        504        3
Unknown                         10                    8     3               1
3              0        2            0      1          1          0        0

================================================================================
  CONVERSION RATES (%):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC   MCZ  E.L.M. LEBLANC
FRISQUET  SAUNIER DUVAL  HITACHI  DE DIETRICH  DOVRE  VIESSMANN  JOLLY MEC
SOL&MOI
main_equipment_category
AC                            34.6                 32.6  19.0            40.0
50.0           66.7     22.0          0.0   50.0        0.0       66.7      0.0
Boiler                        41.4                 32.8  41.7            37.4
45.8           35.4     31.3         45.7   50.0       32.1        NaN      0.0
Heat Pump                     44.9                 37.3  37.8             NaN
30.0           18.8     43.9         31.7   77.8       50.7       50.0     43.8
Other                         34.5                 26.1  35.0            53.9
21.4           47.5     29.3         38.9   27.3       27.8       33.3     23.2
Stove                         23.2                  9.1  49.1             NaN
100.0            NaN     22.2          0.0   41.4       50.0       41.9      0.0
Unknown                       60.0                 62.5  33.3             0.0
100.0            NaN     50.0          NaN    0.0        0.0        NaN      NaN

================================================================================
  BEST BRAND BY PRODUCT TYPE
--------------------------------------------------------------------------------
```

```
Heat Pump:
  Best brand: DOVRE
  Conversion: 77.8%
  Customers: 9
  Top 3 brands:
    1. DOVRE: 77.8% (n=9)
    2. VIESSMANN: 50.7% (n=286)
    3. JOLLY MEC: 50.0% (n=2)

Boiler:
  Best brand: DOVRE
  Conversion: 50.0%
  Customers: 6
  Top 3 brands:
    1. DOVRE: 50.0% (n=6)
    2. FRISQUET: 45.8% (n=1,438)
    3. DE DIETRICH: 45.7% (n=823)

Stove:
  Best brand: FRISQUET
  Conversion: 100.0%
  Customers: 1
  Top 3 brands:
    1. FRISQUET: 100.0% (n=1)
    2. VIESSMANN: 50.0% (n=2)
    3. MCZ: 49.1% (n=2,473)

AC:
  Best brand: JOLLY MEC
  Conversion: 66.7%
  Customers: 3
  Top 3 brands:
    1. JOLLY MEC: 66.7% (n=3)
    2. SAUNIER DUVAL: 66.7% (n=9)
    3. FRISQUET: 50.0% (n=16)

Other:
  Best brand: E.L.M. LEBLANC
  Conversion: 53.9%
  Customers: 297
  Top 3 brands:
    1. E.L.M. LEBLANC: 53.9% (n=297)
    2. SAUNIER DUVAL: 47.5% (n=61)
    3. DE DIETRICH: 38.9% (n=72)

================================================================================
 BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------
```

```
ATLANTIC:
  Best product: Unknown
  Conversion: 60.0%
  Customers: 10
  All products:
    Unknown: 60.0% (n=10)
    Heat Pump: 44.9% (n=1,900)
    Boiler: 41.4% (n=879)
    AC: 34.6% (n=1,111)
    Other: 34.5% (n=504)
    Stove: 23.2% (n=56)

MITSUBISHI ELECTRIC:
  Best product: Unknown
  Conversion: 62.5%
  Customers: 8
  All products:
    Unknown: 62.5% (n=8)
    Heat Pump: 37.3% (n=1,287)
    Boiler: 32.8% (n=119)
    AC: 32.6% (n=2,595)
    Other: 26.1% (n=23)
    Stove: 9.1% (n=11)

MCZ:
  Best product: Stove
  Conversion: 49.1%
  Customers: 2,473
  All products:
    Stove: 49.1% (n=2,473)
    Boiler: 41.7% (n=12)
    Heat Pump: 37.8% (n=45)
    Other: 35.0% (n=20)
    Unknown: 33.3% (n=3)
    AC: 19.0% (n=21)

E.L.M. LEBLANC:
  Best product: Other
  Conversion: 53.9%
  Customers: 297
  All products:
    Other: 53.9% (n=297)
    AC: 40.0% (n=15)
    Boiler: 37.4% (n=1,493)
    Unknown: 0.0% (n=1)

FRISQUET:
```

```
  Best product: Stove
  Conversion: 100.0%
  Customers: 1
  All products:
    Stove: 100.0% (n=1)
    Unknown: 100.0% (n=3)
    AC: 50.0% (n=16)
    Boiler: 45.8% (n=1,438)
    Heat Pump: 30.0% (n=40)
    Other: 21.4% (n=14)

SAUNIER DUVAL:
  Best product: AC
  Conversion: 66.7%
  Customers: 9
  All products:
    AC: 66.7% (n=9)
    Other: 47.5% (n=61)
    Boiler: 35.4% (n=982)
    Heat Pump: 18.8% (n=32)

HITACHI:
  Best product: Unknown
  Conversion: 50.0%
  Customers: 2
  All products:
    Unknown: 50.0% (n=2)
    Heat Pump: 43.9% (n=469)
    Boiler: 31.3% (n=99)
    Other: 29.3% (n=92)
    Stove: 22.2% (n=9)
    AC: 22.0% (n=291)

DE DIETRICH:
  Best product: Boiler
  Conversion: 45.7%
  Customers: 823
  All products:
    Boiler: 45.7% (n=823)
    Other: 38.9% (n=72)
    Heat Pump: 31.7% (n=41)
    AC: 0.0% (n=5)
    Stove: 0.0% (n=3)

================================================================================
 STRATEGIC INSIGHTS
================================================================================
```

```
Brand Dominance by Product Category:

Heat Pump:
  Volume leader: ATLANTIC (1,900 customers, 46.0% share)
  Conversion leader: DOVRE (77.8%)

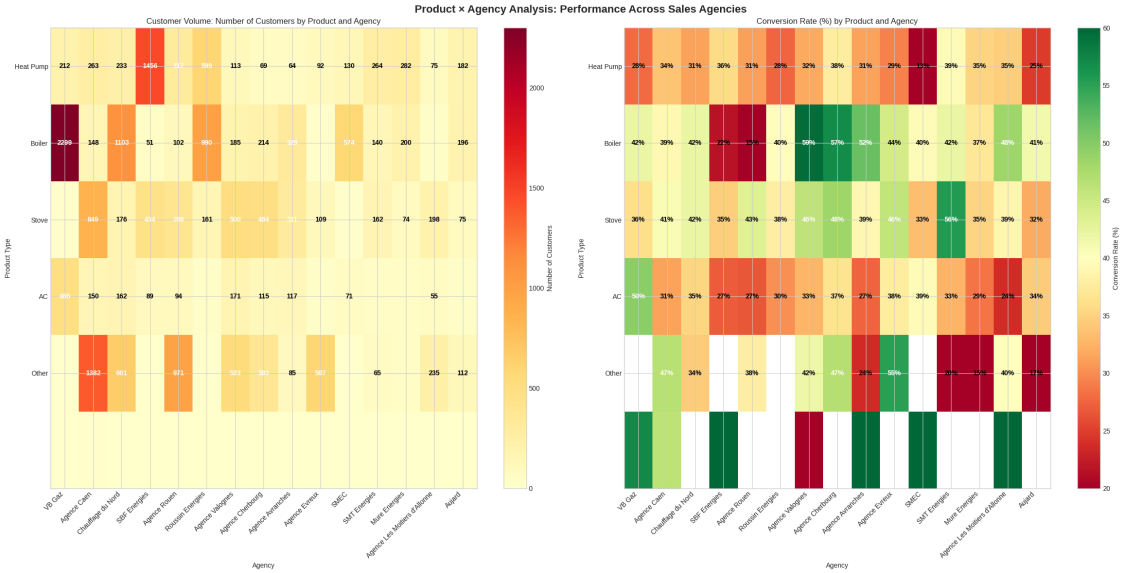
Boiler:
  Volume leader: E.L.M. LEBLANC (1,493 customers, 24.4% share)
  Conversion leader: DOVRE (50.0%)

Stove:
  Volume leader: MCZ (2,473 customers, 67.5% share)
  Conversion leader: FRISQUET (100.0%)

AC:
  Volume leader: MITSUBISHI ELECTRIC (2,595 customers, 63.7% share)
  Conversion leader: SAUNIER DUVAL (66.7%)

Other:
  Volume leader: ATLANTIC (504 customers, 32.7% share)
  Conversion leader: E.L.M. LEBLANC (53.9%)

Top 15 agencies by customer volume:
  VB Gaz: 3,029 customers, 42.3% conversion
  Agence Caen: 2,805 customers, 42.4% conversion
  Chauffage du Nord: 2,335 customers, 38.4% conversion
  SBF Energies: 2,035 customers, 34.7% conversion
  Agence Rouen: 1,880 customers, 36.1% conversion
  Roussin Energies: 1,783 customers, 35.3% conversion
  Agence Valognes: 1,493 customers, 43.6% conversion
  Agence Cherbourg: 1,265 customers, 47.7% conversion
  Agence Avranches: 907 customers, 40.1% conversion
  Agence Evreux: 829 customers, 50.2% conversion
  SMEC: 797 customers, 35.1% conversion
  SMT Energies: 674 customers, 41.5% conversion
  Mure Energies: 645 customers, 34.0% conversion
  Agence Les Moitiers d'Allonne: 595 customers, 38.0% conversion
  Aujard: 594 customers, 30.1% conversion
```




```
================================================================================
PRODUCT × AGENCY ANALYSIS
================================================================================

 CUSTOMER VOLUME BY PRODUCT AND AGENCY:
--------------------------------------------------------------------------------
--------------------
main_agency                VB Gaz  Agence Caen  Chauffage du Nord  SBF Energies
Agence Rouen  Roussin Energies  Agence Valognes  Agence Cherbourg  Agence
Avranches  Agence Evreux  SMEC  SMT Energies  Mure Energies  Agence Les Moitiers
d'Allonne  Aujard
main_equipment_category
AC                          212          263                  233          1456
317                599              113                   69                 64
92   130            264            282                                  75      182
Boiler                     2299          148                 1103            51
102                990              185                  214                329
9   574            140             200                                  31      196
Heat Pump                    25          849                  176           434
396                161              500                  484                311
109    21            162              74                                  198       75
Other                       486          150                  162            89
94                 33               171                  115                117
32    71             43              49                                  55       29
Stove                         0         1382                  661             0
971                 0               523                  383                 85
587     0             65              40                                  235      112
```

```
Unknown                        7           13                   0             5
0                  0                1                  0                  1
0     1             0              0                                1       0

================================================================================
  CONVERSION RATES (%) BY PRODUCT AND AGENCY:
--------------------------------------------------------------------------------
--------------------
main_agency              VB Gaz  Agence Caen  Chauffage du Nord  SBF Energies
Agence Rouen  Roussin Energies  Agence Valognes  Agence Cherbourg  Agence
Avranches  Agence Evreux   SMEC  SMT Energies  Mure Energies  Agence Les
Moitiers d'Allonne  Aujard
main_equipment_category
AC                         27.8         33.8               31.3          35.6
31.2              27.5             31.9              37.7              31.2
29.3   13.1          39.4           35.1                            34.7    24.7
Boiler                     42.1         39.2               42.2          21.6
14.7              39.7             59.5              57.0              51.7
44.4   39.5          42.1           37.0                            48.4    41.3
Heat Pump                  36.0         40.6               42.0          34.6
43.4              37.9             46.0              48.3              38.9
45.9   33.3          55.6           35.1                            38.9    32.0
Other                      49.6         31.3               35.2          27.0
26.6              30.3             32.7              36.5              27.4
37.5   39.4          32.6           28.6                            23.6    34.5
Stove                       NaN         46.7               34.5           NaN
37.9               NaN             41.9              46.7              23.5
55.0    NaN          20.0           15.0                            40.0    17.0
Unknown                    57.1         46.2                NaN          60.0
NaN               NaN              0.0               NaN             100.0
NaN  100.0           NaN            NaN                           100.0     NaN

================================================================================
  BEST AGENCY BY PRODUCT TYPE
--------------------------------------------------------------------------------
--------------------

Heat Pump:
  Best agency: SMT Energies
  Conversion: 55.6%
  Customers: 162
  Top 3 agencies:
    1. SMT Energies: 55.6% (n=162)
    2. Agence Cherbourg: 48.3% (n=484)
    3. Agence Valognes: 46.0% (n=500)

Boiler:
  Best agency: Agence Valognes
```

```
  Conversion: 59.5%
  Customers: 185
  Top 3 agencies:
    1. Agence Valognes: 59.5% (n=185)
    2. Agence Cherbourg: 57.0% (n=214)
    3. Agence Avranches: 51.7% (n=329)

Stove:
  Best agency: Agence Evreux
  Conversion: 55.0%
  Customers: 587
  Top 3 agencies:
    1. Agence Evreux: 55.0% (n=587)
    2. Agence Cherbourg: 46.7% (n=383)
    3. Agence Caen: 46.7% (n=1,382)

AC:
  Best agency: SMT Energies
  Conversion: 39.4%
  Customers: 264
  Top 3 agencies:
    1. SMT Energies: 39.4% (n=264)
    2. Agence Cherbourg: 37.7% (n=69)
    3. SBF Energies: 35.6% (n=1,456)

Other:
  Best agency: VB Gaz
  Conversion: 49.6%
  Customers: 486
  Top 3 agencies:
    1. VB Gaz: 49.6% (n=486)
    2. SMEC: 39.4% (n=71)
    3. Agence Evreux: 37.5% (n=32)

================================================================================
 BEST PRODUCT BY AGENCY
--------------------------------------------------------------------------------
--------------------

VB Gaz:
  Best product: Unknown
  Conversion: 57.1%
  Customers: 7
  All products:
    Unknown: 57.1% (n=7)
    Other: 49.6% (n=486)
    Boiler: 42.1% (n=2,299)
    Heat Pump: 36.0% (n=25)
```

```
    AC: 27.8% (n=212)

Agence Caen:
  Best product: Stove
  Conversion: 46.7%
  Customers: 1,382
  All products:
    Stove: 46.7% (n=1,382)
    Unknown: 46.2% (n=13)
    Heat Pump: 40.6% (n=849)
    Boiler: 39.2% (n=148)
    AC: 33.8% (n=263)
    Other: 31.3% (n=150)

Chauffage du Nord:
  Best product: Boiler
  Conversion: 42.2%
  Customers: 1,103
  All products:
    Boiler: 42.2% (n=1,103)
    Heat Pump: 42.0% (n=176)
    Other: 35.2% (n=162)
    Stove: 34.5% (n=661)
    AC: 31.3% (n=233)

SBF Energies:
  Best product: Unknown
  Conversion: 60.0%
  Customers: 5
  All products:
    Unknown: 60.0% (n=5)
    AC: 35.6% (n=1,456)
    Heat Pump: 34.6% (n=434)
    Other: 27.0% (n=89)
    Boiler: 21.6% (n=51)

Agence Rouen:
  Best product: Heat Pump
  Conversion: 43.4%
  Customers: 396
  All products:
    Heat Pump: 43.4% (n=396)
    Stove: 37.9% (n=971)
    AC: 31.2% (n=317)
    Other: 26.6% (n=94)
    Boiler: 14.7% (n=102)

Roussin Energies:
```

```
  Best product: Boiler
  Conversion: 39.7%
  Customers: 990
  All products:
    Boiler: 39.7% (n=990)
    Heat Pump: 37.9% (n=161)
    Other: 30.3% (n=33)
    AC: 27.5% (n=599)

Agence Valognes:
  Best product: Boiler
  Conversion: 59.5%
  Customers: 185
  All products:
    Boiler: 59.5% (n=185)
    Heat Pump: 46.0% (n=500)
    Stove: 41.9% (n=523)
    Other: 32.7% (n=171)
    AC: 31.9% (n=113)
    Unknown: 0.0% (n=1)

Agence Cherbourg:
  Best product: Boiler
  Conversion: 57.0%
  Customers: 214
  All products:
    Boiler: 57.0% (n=214)
    Heat Pump: 48.3% (n=484)
    Stove: 46.7% (n=383)
    AC: 37.7% (n=69)
    Other: 36.5% (n=115)

Agence Avranches:
  Best product: Unknown
  Conversion: 100.0%
  Customers: 1
  All products:
    Unknown: 100.0% (n=1)
    Boiler: 51.7% (n=329)
    Heat Pump: 38.9% (n=311)
    AC: 31.2% (n=64)
    Other: 27.4% (n=117)
    Stove: 23.5% (n=85)

Agence Evreux:
  Best product: Stove
  Conversion: 55.0%
  Customers: 587
```

```
  All products:
    Stove: 55.0% (n=587)
    Heat Pump: 45.9% (n=109)
    Boiler: 44.4% (n=9)
    Other: 37.5% (n=32)
    AC: 29.3% (n=92)

================================================================================
 AGENCY PERFORMANCE SUMMARY
================================================================================

Top 15 agencies by volume:
                              conversion_rate  customer_count  heat_pump_pct
subsidy_issue_pct  avg_price
main_agency
VB Gaz                                   42.3            3029            1.0
1.0   4820.074
Agence Caen                              42.4            2805           31.7
40.7   8919.356
Chauffage du Nord                        38.4            2335            8.7
34.7   6223.901
SBF Energies                             34.7            2035           24.9
14.5   9368.052
Agence Rouen                             36.1            1880           23.6
45.2   8761.846
Roussin Energies                         35.3            1783           10.5
10.7   7365.723
Agence Valognes                          43.6            1493           36.7
46.4   9873.781
Agence Cherbourg                         47.7            1265           41.4
47.9   9367.784
Agence Avranches                         40.1             907           40.1
37.4  10558.960
Agence Evreux                            50.2             829           13.8
46.6   7437.364
SMEC                                     35.1             797            3.3
6.1   6255.808
SMT Energies                             41.5             674           28.9
38.7   7058.363
Mure Energies                            34.0             645           14.1
13.3   6804.141
Agence Les Moitiers d'Allonne            38.0             595           35.0
40.8  10305.702
Aujard                                   30.1             594           13.5
11.1   8247.407

================================================================================
 STRATEGIC INSIGHTS
```

```
================================================================================

Agency Excellence by Product Category:

Heat Pump:
  Conversion leader: SMT Energies (55.6%, n=162)

Boiler:
  Conversion leader: Agence Valognes (59.5%, n=185)

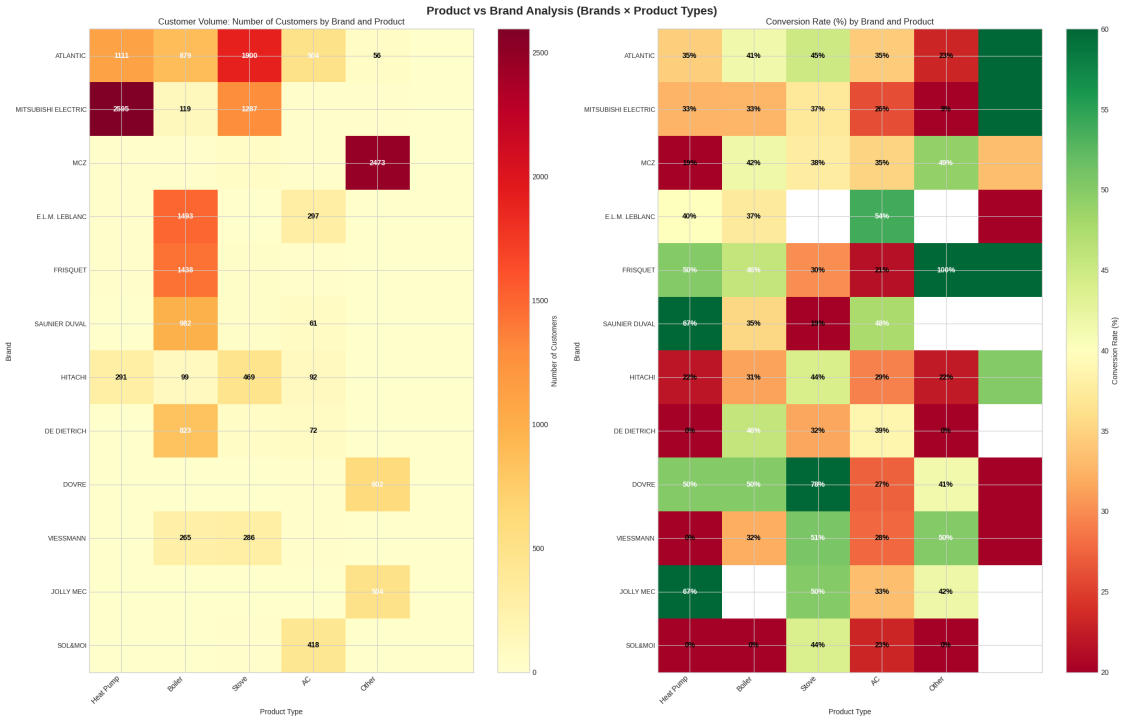
Stove:
  Conversion leader: Agence Evreux (55.0%, n=587)

AC:
  Conversion leader: SMT Energies (39.4%, n=264)

Other:
  Conversion leader: VB Gaz (49.6%, n=486)

================================================================================
 HEAT PUMP SPECIALISTS
================================================================================

Top 5 agencies for Heat Pump conversion:
  SMT Energies: 55.6% (n=162)
  Agence Cherbourg: 48.3% (n=484)
  Agence Valognes: 46.0% (n=500)
  Agence Evreux: 45.9% (n=109)
  Agence Rouen: 43.4% (n=396)
```




```
================================================================================
BRAND × PRODUCT ANALYSIS (Swapped Axes)
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove  Unknown
main_brand
ATLANTIC                 1111     879       1900    504     56       10
MITSUBISHI ELECTRIC      2595     119       1287     23     11        8
MCZ                        21      12         45     20   2473        3
E.L.M. LEBLANC             15    1493          0    297      0        1
FRISQUET                   16    1438         40     14      1        3
SAUNIER DUVAL               9     982         32     61      0        0
HITACHI                   291      99        469     92      9        2
DE DIETRICH                 5     823         41     72      3        0
DOVRE                       2       6          9     11    602        1
VIESSMANN                   3     265        286     18      2        1
JOLLY MEC                   3       0          2      9    504        0
SOL&MOI                     1       2         16    418      3        0

================================================================================
  CONVERSION RATES (%):
```

```
--------------------------------------------------------------------------------
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove  Unknown
main_brand
ATLANTIC                 34.6    41.4       44.9   34.5   23.2     60.0
MITSUBISHI ELECTRIC      32.6    32.8       37.3   26.1    9.1     62.5
MCZ                      19.0    41.7       37.8   35.0   49.1     33.3
E.L.M. LEBLANC           40.0    37.4        NaN   53.9    NaN      0.0
FRISQUET                 50.0    45.8       30.0   21.4  100.0    100.0
SAUNIER DUVAL            66.7    35.4       18.8   47.5    NaN      NaN
HITACHI                  22.0    31.3       43.9   29.3   22.2     50.0
DE DIETRICH               0.0    45.7       31.7   38.9    0.0      NaN
DOVRE                    50.0    50.0       77.8   27.3   41.4      0.0
VIESSMANN                 0.0    32.1       50.7   27.8   50.0      0.0
JOLLY MEC                66.7     NaN       50.0   33.3   41.9      NaN
SOL&MOI                   0.0     0.0       43.8   23.2    0.0      NaN

================================================================================
  BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------

ATLANTIC:
  Best product: Unknown
  Conversion: 60.0%
  Customers: 10
  Top 3 products:
    1. Unknown: 60.0% (n=10)
    2. Heat Pump: 44.9% (n=1,900)
    3. Boiler: 41.4% (n=879)

MITSUBISHI ELECTRIC:
  Best product: Unknown
  Conversion: 62.5%
  Customers: 8
  Top 3 products:
    1. Unknown: 62.5% (n=8)
    2. Heat Pump: 37.3% (n=1,287)
    3. Boiler: 32.8% (n=119)

MCZ:
  Best product: Stove
  Conversion: 49.1%
  Customers: 2,473
  Top 3 products:
    1. Stove: 49.1% (n=2,473)
    2. Boiler: 41.7% (n=12)
    3. Heat Pump: 37.8% (n=45)

E.L.M. LEBLANC:
```

```
  Best product: Other
  Conversion: 53.9%
  Customers: 297
  Top 3 products:
    1. Other: 53.9% (n=297)
    2. AC: 40.0% (n=15)
    3. Boiler: 37.4% (n=1,493)

FRISQUET:
  Best product: Stove
  Conversion: 100.0%
  Customers: 1
  Top 3 products:
    1. Stove: 100.0% (n=1)
    2. Unknown: 100.0% (n=3)
    3. AC: 50.0% (n=16)

SAUNIER DUVAL:
  Best product: AC
  Conversion: 66.7%
  Customers: 9
  Top 3 products:
    1. AC: 66.7% (n=9)
    2. Other: 47.5% (n=61)
    3. Boiler: 35.4% (n=982)

HITACHI:
  Best product: Unknown
  Conversion: 50.0%
  Customers: 2
  Top 3 products:
    1. Unknown: 50.0% (n=2)
    2. Heat Pump: 43.9% (n=469)
    3. Boiler: 31.3% (n=99)

DE DIETRICH:
  Best product: Boiler
  Conversion: 45.7%
  Customers: 823
  Top 3 products:
    1. Boiler: 45.7% (n=823)
    2. Other: 38.9% (n=72)
    3. Heat Pump: 31.7% (n=41)

DOVRE:
  Best product: Heat Pump
  Conversion: 77.8%
  Customers: 9
```

```
  Top 3 products:
    1. Heat Pump: 77.8% (n=9)
    2. AC: 50.0% (n=2)
    3. Boiler: 50.0% (n=6)

VIESSMANN:
  Best product: Heat Pump
  Conversion: 50.7%
  Customers: 286
  Top 3 products:
    1. Heat Pump: 50.7% (n=286)
    2. Stove: 50.0% (n=2)
    3. Boiler: 32.1% (n=265)

================================================================================
 BEST BRAND BY PRODUCT TYPE
--------------------------------------------------------------------------------

Heat Pump:
  Best brand: DOVRE
  Conversion: 77.8%
  Customers: 9
  Top 5 brands:
    1. DOVRE: 77.8% (n=9)
    2. VIESSMANN: 50.7% (n=286)
    3. JOLLY MEC: 50.0% (n=2)
    4. ATLANTIC: 44.9% (n=1,900)
    5. HITACHI: 43.9% (n=469)

Boiler:
  Best brand: DOVRE
  Conversion: 50.0%
  Customers: 6
  Top 5 brands:
    1. DOVRE: 50.0% (n=6)
    2. FRISQUET: 45.8% (n=1,438)
    3. DE DIETRICH: 45.7% (n=823)
    4. MCZ: 41.7% (n=12)
    5. ATLANTIC: 41.4% (n=879)

Stove:
  Best brand: FRISQUET
  Conversion: 100.0%
  Customers: 1
  Top 5 brands:
    1. FRISQUET: 100.0% (n=1)
    2. VIESSMANN: 50.0% (n=2)
    3. MCZ: 49.1% (n=2,473)
```

```
    4. JOLLY MEC: 41.9% (n=504)
    5. DOVRE: 41.4% (n=602)

AC:
  Best brand: JOLLY MEC
  Conversion: 66.7%
  Customers: 3
  Top 5 brands:
    1. JOLLY MEC: 66.7% (n=3)
    2. SAUNIER DUVAL: 66.7% (n=9)
    3. FRISQUET: 50.0% (n=16)
    4. DOVRE: 50.0% (n=2)
    5. E.L.M. LEBLANC: 40.0% (n=15)

Other:
  Best brand: E.L.M. LEBLANC
  Conversion: 53.9%
  Customers: 297
  Top 5 brands:
    1. E.L.M. LEBLANC: 53.9% (n=297)
    2. SAUNIER DUVAL: 47.5% (n=61)
    3. DE DIETRICH: 38.9% (n=72)
    4. MCZ: 35.0% (n=20)
    5. ATLANTIC: 34.5% (n=504)

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Brand Performance by Product Category:

ATLANTIC:
  Unknown: 60.0% (n=10)
  Heat Pump: 44.9% (n=1,900)
  Boiler: 41.4% (n=879)

MITSUBISHI ELECTRIC:
  Unknown: 62.5% (n=8)
  Heat Pump: 37.3% (n=1,287)
  Boiler: 32.8% (n=119)

MCZ:
  Stove: 49.1% (n=2,473)
  Boiler: 41.7% (n=12)
  Heat Pump: 37.8% (n=45)

E.L.M. LEBLANC:
  Other: 53.9% (n=297)
```

```
  AC: 40.0% (n=15)
  Boiler: 37.4% (n=1,493)

FRISQUET:
  Stove: 100.0% (n=1)
  Unknown: 100.0% (n=3)
  AC: 50.0% (n=16)

SAUNIER DUVAL:
  AC: 66.7% (n=9)
  Other: 47.5% (n=61)
  Boiler: 35.4% (n=982)

HITACHI:
  Unknown: 50.0% (n=2)
  Heat Pump: 43.9% (n=469)
  Boiler: 31.3% (n=99)

DE DIETRICH:
  Boiler: 45.7% (n=823)
  Other: 38.9% (n=72)
  Heat Pump: 31.7% (n=41)

================================================================================
 MARKET LEADERS BY PRODUCT CATEGORY
================================================================================

Heat Pump:
  Volume leader: ATLANTIC (1,900 customers, 46.0% share)
  Conversion leader: DOVRE (77.8%)

Boiler:
  Volume leader: E.L.M. LEBLANC (1,493 customers, 24.4% share)
  Conversion leader: DOVRE (50.0%)

Stove:
  Volume leader: MCZ (2,473 customers, 67.5% share)
  Conversion leader: FRISQUET (100.0%)

AC:
  Volume leader: MITSUBISHI ELECTRIC (2,595 customers, 63.7% share)
  Conversion leader: SAUNIER DUVAL (66.7%)

Other:
  Volume leader: ATLANTIC (504 customers, 32.7% share)
  Conversion leader: E.L.M. LEBLANC (53.9%)
```

```
(main_equipment_category    AC  Boiler  Heat Pump  Other  Stove  Unknown
 main_brand
 ATLANTIC                 1111     879       1900    504     56       10
 MITSUBISHI ELECTRIC      2595     119       1287     23     11        8
 MCZ                        21      12         45     20   2473        3
 E.L.M. LEBLANC             15    1493          0    297      0        1
 FRISQUET                   16    1438         40     14      1        3
 SAUNIER DUVAL               9     982         32     61      0        0
 HITACHI                   291      99        469     92      9        2
 DE DIETRICH                 5     823         41     72      3        0
 DOVRE                       2       6          9     11    602        1
 VIESSMANN                   3     265        286     18      2        1
 JOLLY MEC                   3       0          2      9    504        0
 SOL&MOI                     1       2         16    418      3        0,
 main_equipment_category         AC     Boiler  Heat Pump      Other  \
 main_brand
 ATLANTIC                 34.563456  41.410694  44.894737  34.523810
 MITSUBISHI ELECTRIC      32.601156  32.773109  37.296037  26.086957
 MCZ                      19.047619  41.666667  37.777778  35.000000
 E.L.M. LEBLANC           40.000000  37.441393        NaN  53.872054
 FRISQUET                 50.000000  45.757997  30.000000  21.428571
 SAUNIER DUVAL            66.666667  35.437882  18.750000  47.540984
 HITACHI                  21.993127  31.313131  43.923241  29.347826
 DE DIETRICH               0.000000  45.686513  31.707317  38.888889
 DOVRE                    50.000000  50.000000  77.777778  27.272727
 VIESSMANN                 0.000000  32.075472  50.699301  27.777778
 JOLLY MEC                66.666667        NaN  50.000000  33.333333
 SOL&MOI                   0.000000   0.000000  43.750000  23.205742

 main_equipment_category       Stove     Unknown
 main_brand
 ATLANTIC                  23.214286   60.000000
 MITSUBISHI ELECTRIC        9.090909   62.500000
 MCZ                       49.090174   33.333333
 E.L.M. LEBLANC                  NaN    0.000000
 FRISQUET                 100.000000  100.000000
 SAUNIER DUVAL                   NaN         NaN
 HITACHI                   22.222222   50.000000
 DE DIETRICH                0.000000         NaN
 DOVRE                     41.362126    0.000000
 VIESSMANN                 50.000000    0.000000
 JOLLY MEC                 41.865079         NaN
 SOL&MOI                    0.000000         NaN  )
```

```python
import ipynbname
import subprocess
import os
```

```python
import sys

def export_notebook_to_pdf(overwrite=True, verbose=True):
    """
    Export current notebook to PDF with overwrite control

    Args:
        overwrite (bool): If True, overwrite existing PDF file
        verbose (bool): If True, print progress messages

    Returns:
        bool: True if successful, False otherwise
    """
    try:
        # Get notebook path
        notebook_path = ipynbname.path()
        notebook_path_str = str(notebook_path)
        pdf_path = notebook_path_str.replace('.ipynb', '.pdf')

        if verbose:
            print(f"  Notebook: {notebook_path.name}")
            print(f"  PDF path: {pdf_path}")

        # Check if PDF exists
        if os.path.exists(pdf_path):
            if overwrite:
                if verbose:
                    print(f"  Removing existing PDF: {pdf_path}")
                os.remove(pdf_path)
            else:
                if verbose:
                    print(f"  PDF already exists and overwrite=False")
                return False

        # Convert to PDF
        if verbose:
            print("  Converting to PDF...")

        result = subprocess.run([
            'jupyter', 'nbconvert', '--to', 'pdf',
            '--no-prompt',  # Suppress input prompts
            notebook_path_str
        ], capture_output=True, text=True)

        if result.returncode == 0:
            if verbose:
                print(f"  PDF saved: {pdf_path}")
```

```
            return True
        else:
            if verbose:
                print(f"  Conversion failed:")
                print(result.stderr)
            return False

    except Exception as e:
        if verbose:
            print(f"  Error: {e}")
        return False

# Use the function
if __name__ == "__main__":
    # Overwrite existing PDF
    export_notebook_to_pdf(overwrite=True)

    # Or create new version without overwriting
    # export_notebook_to_pdf(overwrite=False)export_full_notebook_to_pdf()
```

```
Notebook: _eda_price.ipynb
PDF path: /home/valeriya/project/homeserve/ai-france-hvac/_eda_price.pdf
Converting to PDF…
PDF saved: /home/valeriya/project/homeserve/ai-france-hvac/_eda_price.pdf
```